CS 310 : Automata Theory

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jp(L) and sc(L)

Let L be a regular language.

- jp(L) is the smallest n satisfying Jaffe's pumping property for L
- sc(L) is the size of the minimal DFA for L
- In general : $\text{jp}(L) \leq \text{sc}(L)$ (why?)

Let $n = \text{sc}(L)$. Take any word $y = a_1 \ldots a_n \in \Sigma^n$. on its run $q_0 \ldots q_n$, y repeats a state: $q_i = q_j$ for some $0 \approx i < j \approx n$. Then choosing $v = a_{i+1} \ldots a_j$ gives a safe Jaffe split for $y = uvw$.

- So words of length n have a safe Jaffe split.
- Thus, $\text{jp}(L) \leq n = \text{sc}(L)$.

In general

For every $k \geq 1$: $jp(L) \leq k$ if and only if every length-k word with a repeat-free run in the minimal DFA admits a safe Jaffe split.

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- So *every* word of length k, repeat-free run or not, has a safe split.
- That means k satisfies Jaffe's property.
- Hence $jp(L) \leq k$.

In general : converse

For every $k \geq 1$: $jp(L) \leq k$ if and only if every length-k word with a repeat-free run in the minimal DFA admits a safe Jaffe split.

Assume $jp(L) \leq k$.

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- Jaffe's property is monotone: once some length k' satisfies it, every longer length also does.
- Since $k \geq jp(L)$, length k satisfies Jaffe's property.
- In particular, *every* word of length k has a safe split, including every one whose run happens to be repeat-free.

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- Keep the *same* u, v, and set $w' := wc$.

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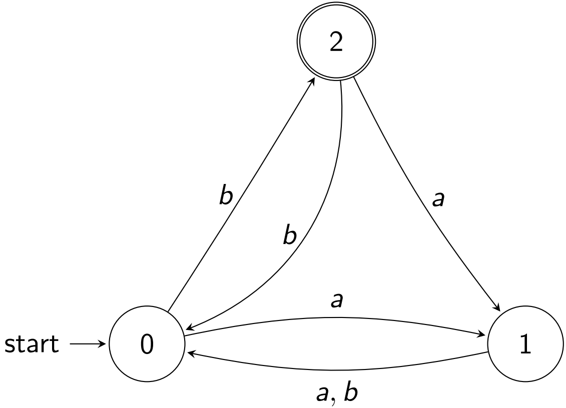
- So u, v, wc *is* a safe split of y'.

Example : $\mathrm{jp}(L) = 2 < \mathrm{sc}(L) = 3$



The word $y = ba$ has a run $0 \to 2 \to 1$: all three states, no repeat.

$\mathsf{jp}(L) = 2 < 3 = \mathsf{sc}(L) = 3$

Split $y = ba$ as $u = \varepsilon,\ v = b,\ w = a$

- $\delta^*(0, v^i) \in \{0, 2\}$ for all i
- all continuations from both 0 and 2 on $w = a$ reach same state 1.
- Safe split

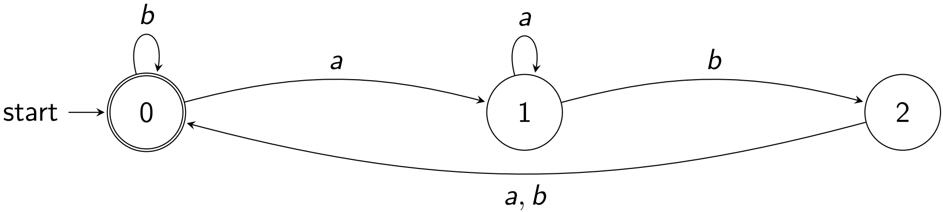
Remaining words aa, ab, bb witness a run with state repetition.

- $y = aa$, run is $0 \to 1 \to 0$. Choose $u = w = \epsilon, v = aa$ as the Jaffe split.
- Likewise for $y = ab, bb$, choose $v = y$ as the Jaffe split.

So all words in Σ^2 have a Jaffe split.

Example : $\mathrm{jp}(L) = 3 = \mathrm{sc}(L)$

Why $\mathrm{jp}(L) > 2$?



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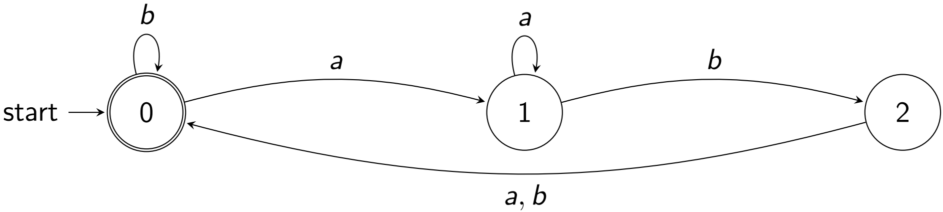

The word $y = ab$ has run $0 \to 1 \to 2$: repeat-free. Checking *every* possible split:

- $u = \varepsilon, v = a, w = b$: $uv^i = a^i$ visits $0, 1, 1, 1, \ldots$.
- Applying $w = b$: $\delta(0, b) = 0$ but $\delta(1, b) = 2$, not a safe split

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- Try $u = \varepsilon,\ v = ab,\ w = \varepsilon$.
- Then $uv^i = (ab)^i$ visits $0 \to 2 \to 0 \to 2 \to \cdots$
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- Try $u = a,\ v = b,\ w = \varepsilon$.
- Then $uv^i = ab^i$ visits $1 \to 2 \to 0 \to 0 \to 0 \to \cdots$
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Check that $\mathrm{jp}(L) = 3 = \mathrm{sc}(L)$

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- for each breakup of w into three words $xyz = w$ such that $y \neq \epsilon$ and $|xy| \leq p$ then
- there is a $k \geq 0$ such that $uxy^kzv \notin L$.

The positive form

- Taking the contrapositive gives the form we actually use to *test* a language:

If L is regular, then for each p: for all u, w, v with $uwv \in L$ and $|w| \geq p$, there is a breakup $w = xyz$ ($y \neq \varepsilon$, $|xy| \approx p$) such that $uxy^kzv \in L$ for *every* $k \geq 0$.

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- Stronger than the classical pumping lemma: the window w can sit *anywhere* inside the string, not just as a bounded prefix.
- Weaker than an unrestricted split, though: once w is fixed, the pump x, y is still confined to the first p symbols of w (exactly as in the classical lemma).

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- We know: regular $\implies$ subword-pumpable.
- Question: does the converse hold? Is every subword-pumpable language also regular?
- A student (call him/her as B) claims : **no**, there is a non-regular language that is still subword-pumpable.
- If B's claim is correct, then we can conclude that the subword pumping lemma is not an iff condition for regularity.

B's construction: $pad(L)$

Let L be any non-regular language over Σ with $\varepsilon \notin L$, and let $\star \notin \Sigma$.

$$pad(L) := \bigcup_{n \geq 1} \left\{ a_1 \star^{i_1} \cdots a_n \star^{i_n} \;:\; a_1, \ldots, a_n \in \Sigma, \right.$$
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- If *any* block of stars deviates from count 1, the word is in $pad(L)$
- Only $a_1 \star a_2 \star \cdots a_n \star$ is constrained: it is in $pad(L)$ exactly when $a_1 \cdots a_n \in L$.

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- Contradiction : L was assumed non-regular. So $pad(L)$ is non-regular.

B's claim : we can pump stars choosing $p = 2$

- B's claim: $pad(L)$ satisfies subword pumping with $p = 2$, since we can just pump a star obtaining words in the language.
- That is, choosing $p = 2$, $\forall u, w, v$ with $uwv \in pad(L)$, and $|w| \geq 2$, there is a split of $w = xyz$, with $y \neq \epsilon$ and $|xy| \leq 2$ such that $uxy^k zv \in pad(L)$ for all $k \geq 0$.
- If you want to check this, don't scroll below

$p = 2$ does not work

- To refute p, we need :
- $\exists\, u, w, v$ (with $uwv \in pad(L)$, $|w| \geq 2$) such that *every* breakup $w = xyz$ ($y \neq \varepsilon$, $|xy| \leq 2$) has *some* k with $uxy^kzv \notin pad(L)$.

Where it breaks: a counterexample

Take $L = \{ab\}$ (any non-regular L would expose the same bug; a finite L is enough here).

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- Choose $w = b\star$ at the very front ($u = \varepsilon$, $v = \star a \star b \star$).
- $|w| = 2 = p$, so every breakup $w = xyz$ automatically satisfies $|xy| \leq p$.

Checking all three breakups of $w = b\star$

(1) $y = \star$, $x = b$, $z = \varepsilon$: $uxy^0zv = \varepsilon \cdot b \cdot \varepsilon \cdot \varepsilon \cdot (\star a \star b\star) = b \star a \star b\star$

- All star counts are 1. But $bab \notin L$. So pumped word not in $pad(L)$.

(2) $y = b$, $x = \varepsilon$, $z = \star$: $uxy^0zv = \varepsilon \cdot \varepsilon \cdot \varepsilon \cdot \star \cdot (\star a \star b\star) = \star \star a \star b\star$ starts with $\star$, not in $pad(L)$ (every word in $pad(L)$ starts with a Σ-letter), not in $pad(L)$.

(3) $y = b\star$, $x = z = \varepsilon$: $uxy^0zv = \varepsilon \cdot \varepsilon \cdot \varepsilon \cdot \varepsilon \cdot (\star a \star b\star) = \star a \star b\star$ again starts with $\star$, not in $pad(L)$.

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All three breakups fail at $k = 0$ and $w = b\star$ has only these three possible breakups, so *every* breakup fails. So $p = 2$ does *not* work for $pad(L)$.

Rescuing B's example

Can we somehow show B's example satisfies subword pumping?

A Suggestion

Let $\Sigma = \{a, b, c\}$ and define

Words containing a non-empty square

$L_{sq} = \{uvvw : u, w \in \Sigma^*,\ v \in \Sigma^+\}$

- Can we show L_{sq} is not regular?
- Perhaps use closure properties to show this?
- Does L_{sq} satisfy subword pumping with $p = 3$?